Global Electric Vehicle Sales Analysis (2010 – 2023)

Using Microsoft Excel

Problem Statement

- The global automotive is rapidly transitioning toward electric mobility, but understanding how EV adoption is evolving across regions and technologies requires proper data analysis.
- The key challenge is to:
  - Identify leading EV markets
  - Analyze EV growth trends
  - Understand which technologies dominate the EV market
  - Explore how EV adoption varies across vehicle types
- This project aims to answer these questions through data analysis and visualization.

Project Overview

- Electric vehicles are becoming an important part of the global transition toward sustainable transportation.
- In this project, I analyzed global EV sales data to understand:
  - Growth of EV adoption over time
  - Leading countries in EV sales
  - Distribution of EV technologies
  - EV adoption across vehicle types
- Using Microsoft Excel, I created an interactive dashboard that visualizes EV market trends from 2010 to 2023.

Project Objectives

- Analyze global EV sales trends from 2010 to 2023
- Identify the top EV markets globally
- Understand the distribution of EV powertrain technologies
- Explore EV sales across different vehicle types
- Build an interactive Excel dashboard to visualize insights

Dataset Description

- The dataset used in this project contains global EV market statistics including:
  - EV sales
  - EV stock
  - EV sales share
  - Powertrain types (BEV, PHEV)
  - Vehicle types (cars, buses, trucks, vans)
  - Region and country data
  - Yearly EV market information
- The dataset contains 12,654 records across multiple countries and years.
- Source: Kaggle (Global EV Data based on International Energy Agency Statistics)

Data Cleaning & Preparation

- Before analysis, the dataset required several preprocessing steps:
  - Filtered relevant parameters such as EV sales, EV stock, and EV sales share
  - Removed future forecast years (2025, 2030, 2035) to focus on historical data
  - Verified numerical values and ensured consistent data formatting
  - Structured the dataset to support Pivot Table analysis
- These steps ensured the dataset was clean and ready for analysis

Tools & Techniques Used

- This project was completed using Microsoft Excel
- Key features used include:
  - Data Cleaning & Preparation
  - Pivot Tables & Charts
  - KPI Calculations
  - Donut Chart
  - Line Charts for trend analysis
  - Bar Charts for comparison
  - TreeMap visualization
  - Interactive slicers for filtering

Who this Dashboard is for ?

- This dashboard can be useful for:
  - Automotive companies analyzing EV markets
  - Policy makers planning sustainable transport policies
  - Investors studying EV market growth
  - Market analysts researching electric mobility trends

Key Performance Indicators (KPIs)

- Total EV Sales (2010 – 2023): $ 247.1 Million
- Total EV Stock: 36 Million Evs
- Largest EV Market: China
- Number of Regions Analyzed: 50
- EV Sales in 2023: $ 92 Million

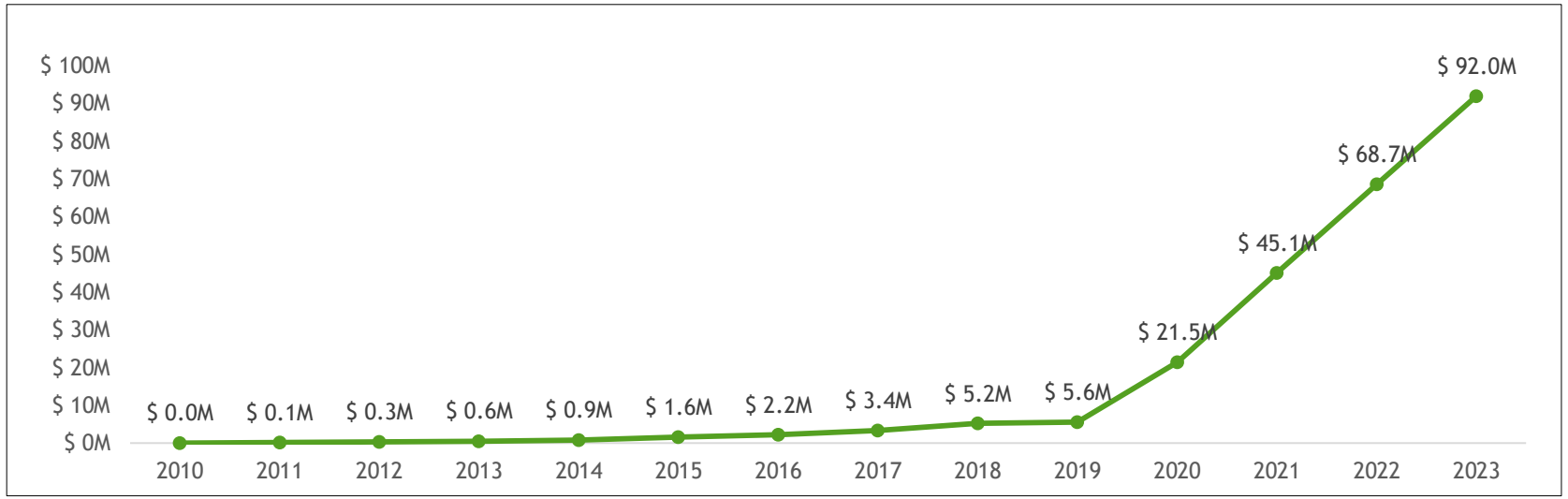
- These KPIs provide a quick summary of the global EV market.

Dashboard Overview

- The dashboard visualizes multiple aspects of the EV market including:
  - Global EV sales trend over time
  - Top EV markets by total sales
  - EV market share growth
  - Distribution of EV powertrain technologies
  - EV sales by vehicle type

- Interactive slicers allow users to filter the dashboard by year and region.

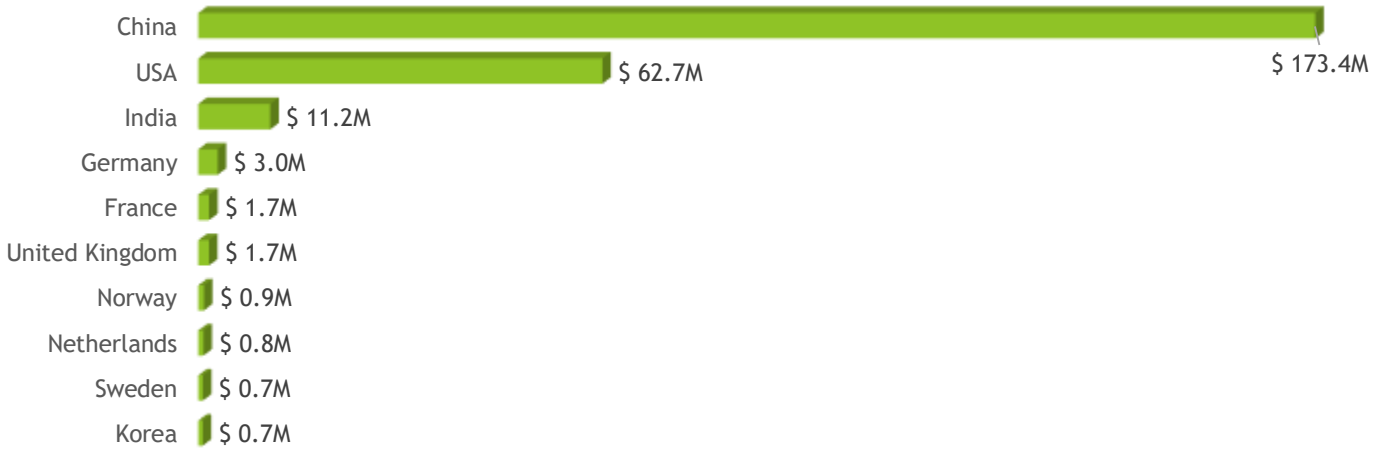
Global EV Sales Trend



- The dashboard shows that global EV sales have grown dramatically since 2010
- Key Observations:
  - EV sales were almost negligible in early years
  - Growth accelerated significantly after 2019
  - EV sales reached approximately $ 92 million in 2023

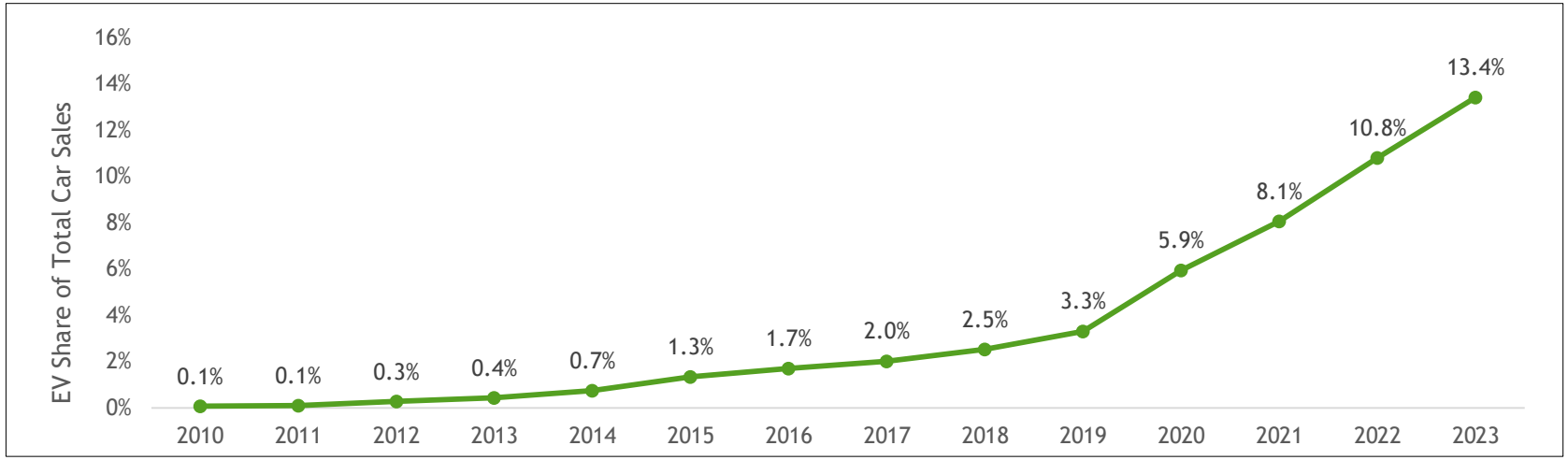
- This indicates strong global adoption of electric vehicles.

Top EV Markets



- The analysis shows the countries with the highest EV sales.
- Key findings:
  - China is the largest EV market globally
  - The United States is the second largest
  - Countries such as Germany, France, and Norway also show strong EV adoption
- This highlights regional differences in EV adoption.

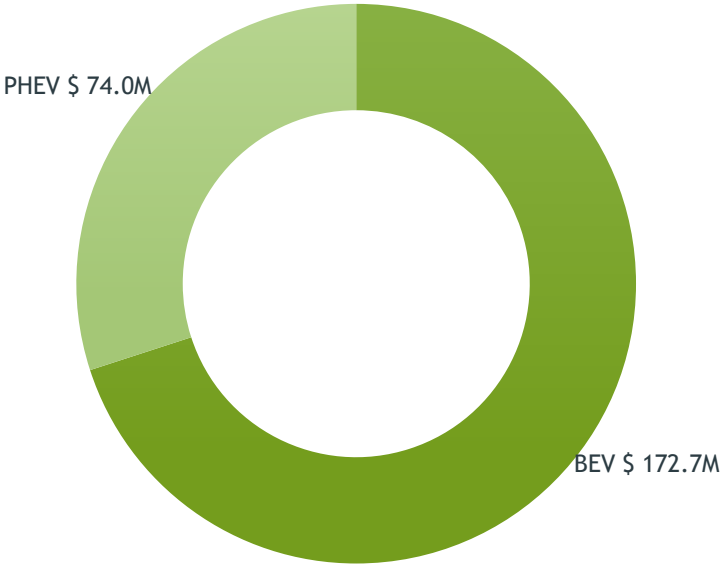
Global EV Market Share Trend



- The EV share of total car sales has increased steadily over time
- Key insights:
  - EV market share was less than 1% before 2014
  - It grew steadily between 2015 and 2019
  - Rapid growth occurred after 2020
  - EV share reached around 13.4% in 2023

- This reflects accelerating global electrification.

EV Powertrain Distribution



- Evs mainly consist of two technologies:
  - Battery Electric Vehicles (BEV)
  - Plug-in Hybrid Electric Vehicles (PHEV)
- From the analysis:
  - BEVs dominate the EV market
  - PHEVs represent a smaller portion of total EV sales
- This suggests that the industry is increasingly shifting toward fully electric vehicles.

EV Sales by Vehicle Type

- The analysis also show EV adoption across vehicle types:
  - Passenger cars dominate EV sales
  - Buses, trucks, and vans represent smaller shares

- This indicates that EV adoption is currently strongest in passenger transportation.

Key Insights

- From the analysis, several key insights emerged:
  - Global EV adoption has accelerated rapidly after 2019
  - China dominates the global EV market
  - Battery Electric Vehicles are the most widely adopted technology
  - Passenger cars account for the majority of EV sales
  - EV market share continues to grow each year

Business Impact

- This dashboard enables stakeholders to:
  - Understand global EV market trends
  - Identify leading EV markets
  - Analyze technology adoption patterns
  - Explore EV adoption across vehicle types
- These insights can support data-driven decisions in the automotive and energy sectors.

Limitations

- Some limitations of this project include:
  - Limited historical data before 2010
  - Some forecast values in the dataset
  - Charging infrastructure data was not included
  - Market share calculations depend on available global car sales data

Future Improvements

- Possible improvements include:
  - Adding EV sales forecasting models
  - Integrating EV charging infrastructure data
  - Creating a more advanced dashboard using PowerBI
  - Adding country-level trend comparisons

Conclusion

- This project demonstrates how Excel can be used to analyze real-world datasets and build interactive dashboards.
- Through this analysis, we explored global EV adoption trends and identified key patterns in EV sales, market share, and technology adoption.
- The project highlights the importance of data-driven insights for understanding the future of electric mobility.